

MTE 203 – Advanced Calculus

Week 3 Tutorial Part 2

Problem 1:

A particle moves along the curve represented by the curve of intersection of the surfaces:

$$z = x^2 + y^2, \quad x + z = 1$$

- Find the equation for the projection of the curve in the x-y plane and draw the curve **[this part was solved during Lecture 4 – see LEARN]**
- Parametrize the curve of intersection so that it is directed counter-clockwise looking from above.
- Setup but do not evaluate the integral to find its length
- Calculate the unit tangent $\hat{T}$ to the curve at the point $(-\frac{1}{2}, \frac{\sqrt{5}}{2}, \frac{3}{2})$
- Calculate the curvature at the point $(-\frac{1}{2}, \frac{\sqrt{5}}{2}, \frac{3}{2})$

Problem 2 [S.11.12 Prob 26]:

- Calculate $\hat{T}, \hat{N}, \hat{B}$ for the curve given by the following parametric equations:

$$x = \cos t,$$

$$y = \sin t,$$

$$z = t.$$

- Express the vector $\vec{F} = x\hat{i} + xy^2\hat{j} + \hat{k}$ in terms of $\hat{T}, \hat{N}$ and $\hat{B}$.